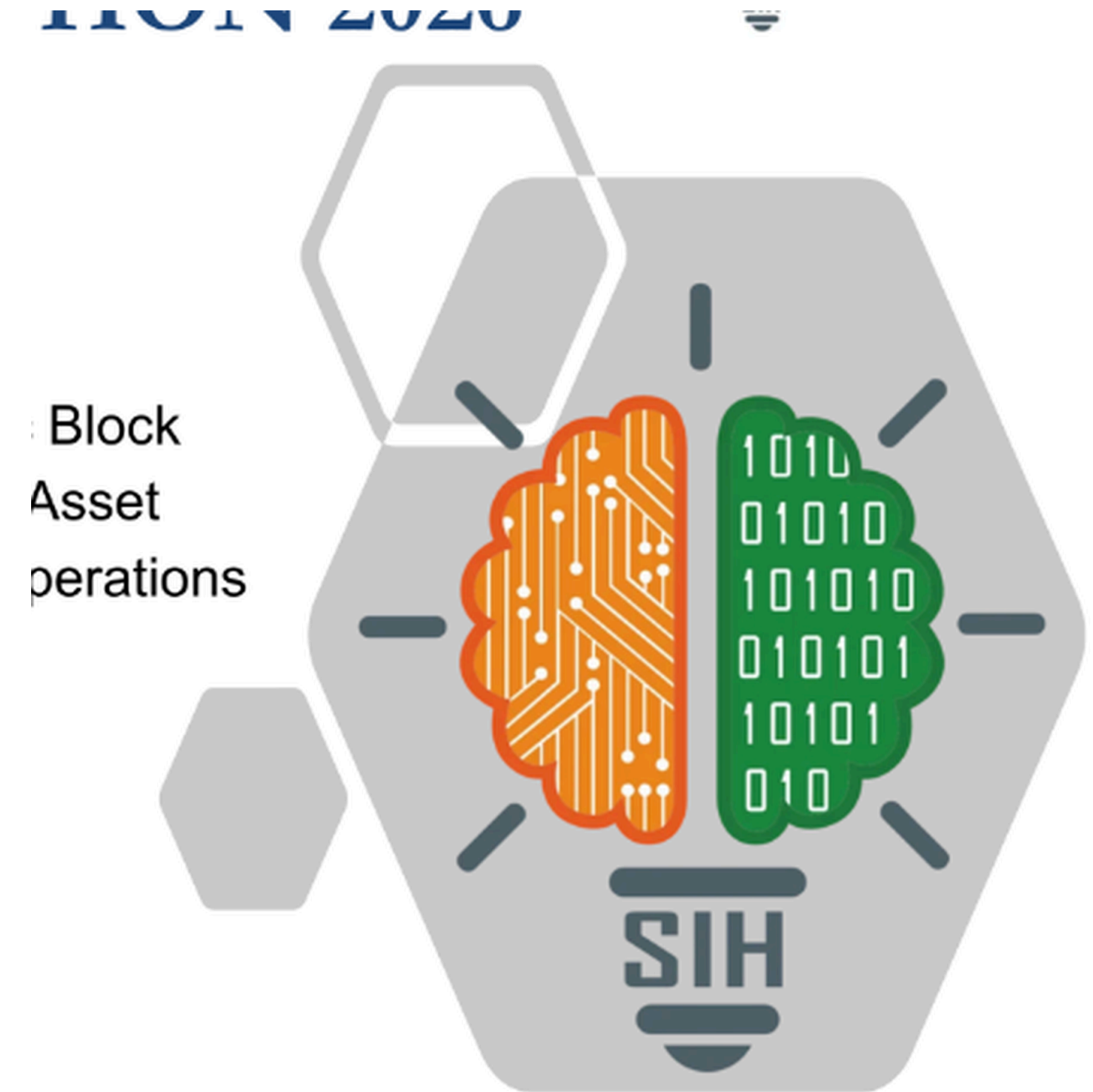


# SMART INDIA HACKATHON 2026

- **Problem Statement ID :** SIH26027
- **Problem Statement Title :** AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways
- **Theme:** Transportation & Logistics
- **Category:** Software
- **Team ID :** NST
- **Team Name :** RailBlock-AI





## Detailed Explanation

- AI Master Coordinator for track maintenance & dispatching.
- Combines **P-Way, TRD & S&T** into 1 joint block window.
- Auto-sides slow trains on loop lines to prevent delays.

## How it Addresses the Problem

- Reduces 4 track closures into 1 joint daily block.
- Isolates train delays to single loop lines.
- Recovers 42.9% lost track capacity on heavy corridors.

## Innovation & Uniqueness

- Google OR-Tools **MILP core** for joint window clustering.
- Dynamic speed & overtake predictor engine.
- Live station-by-station track dashboard visualizer.

### RailBlock-AI: Filling the Gaps in Existing Railway Operations

#### Existing Systems (Missing)

- Siloed Dept Requests (Separate shutdowns for P-Way, TRD, S&T)
- 3-4 Track Shutdowns daily per section
- Manual Loop Siding phone calls
- Cascading Train Delays

#### RailBlock-AI (Included)

- Single-Window Clustering into 1 joint block
- 1 Consolidated Joint Block per section
- Automated AI Siding Directive
- Zero Domino Delays & 42.9% Capacity Recovered



## TECHNOLOGY USED AND METHODOLOGY

| Feature               | Input                                         | Output                                      | How We Do It (Methodology)                                                                       | APIs / Open-Source Tools        |
|-----------------------|-----------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------|
| Joint Block Optimizer | P-Way, TRD & S&T maintenance requests         | 1 Consolidated 4h Joint Maintenance Window  | MILP constraint solver algorithm grouping overlapping track section requests across departments. | Google OR-Tools, PuLP, Python   |
| AI Train Dispatcher   | Live train speeds, priority & telemetry feeds | Loop Line Overtake Directive (PWL Siding)   | Spatio-temporal graph modeling & priority speed vector analysis to hold slow freight trains.     | FastAPI, NetworkX, Redis        |
| Track Capacity Engine | Section downtime logs & historical timetable  | 42.9% Capacity Recovery & Zero Domino Delay | Dynamic track availability modeling & section uptime calculation formulas.                       | Pandas, NumPy, SciPy            |
| Live Route Dashboard  | Route Selection (NDLS → GWL)                  | Live 'Where Is My Track' Interactive Portal | Real-time web portal rendering section tracks, loop siding status & timetable alerts.            | HTML5, Tailwind CSS, Chart.js   |
| CRIS System Gateway   | COA & BDMS railway system feeds               | Automated Controller Approval Directive     | REST API synchronization with Human-in-the-Loop controller click approval workflow.              | CRIS Open APIs, WebSocket, JSON |



# FEASIBILITY AND VIABILITY

## Implementation Feasibility, Operational Risks & Mitigation Strategies

### Feasibility Analysis

- **100% Software:** Zero track hardware changes needed.
- **High Scalability:** Covers all 68 railway divisions.
- **Low Latency:** MILP solver computes in seconds.

### Challenges & Risks

- **Siloed Depts:** Coordinating P-Way, TRD & S&T.
- **Track Speeds:** Handling varying section limits.
- **Change Management:** Shifting controller workflows.

### Strategies to Overcome

- **CRIS Sync:** Direct API hooks to COA & BDMS.
- **Human Control:** Controllers retain click-approval.
- **Phased Rollout:** Starting on Delhi-Mathura route.

#### FEASIBILITY & DEPLOYMENT MATRIX

##### 1. 100% Software Solution

Requires zero track sensors or train hardware modifications. Deploys directly on central CRIS server clusters.

##### 2. High Network Scalability

Modular API architecture easily scales across all 68 Indian Railway divisions & 18 zones nationwide.

##### 3. CRIS System Integration

Direct REST API synchronization with COA & BDMS system feeds with Human-in-the-Loop controller approval.



# IMPACT AND BENEFITS

Operational, Economic, Social & Environmental Value Creation

**42.9%**

## Track Capacity Reclaimed

Consolidates multi-departmental P-Way, TRD & S&T maintenance requests into 1 single joint daily block window.

**Rs. 100s Cr**

## Annual Economic Savings

Prevents locomotive idle hours, reduces diesel freight detention costs, and eliminates wasted track downtime.

**Zero Delay**

## Passenger Punctuality

Isolates slow train holds to station loop lines, allowing fast express passenger trains to run at full speed.

**NRP 2030**

## PM Gati Shakti Scalability

Cloud-native AI brain designed for seamless deployment across all 68 Indian Railway divisions nationwide.



# RESEARCH AND REFERENCES

Academic Papers, Railway Manuals & System Benchmarks

## Government & Policy Reports

- **Ministry of Railways (NRP 2030):** National Rail Plan Capacity Guidelines
- **IRPWM Manual:** Indian Railways Permanent Way Block Protocols

## AI & Railway Research

- **Google OR-Tools:** Constraint Solver for Train & Track Block Scheduling
- **IEEE Transactions (2023):** Dynamic Train Rescheduling & Block Planning

## Existing Systems & Case Studies

- **CRIS COA System:** Control Office Application Live Dispatch Feeds
- **CRIS BDMS Portal:** Block Demand Management System Workflows

